Domain & Range

What can go in? What can come out?

The big idea

A function is a machine: you put in an x, it gives back one y.

- The **domain** is every x you are **allowed to put in**. (inputs, the horizontal direction)
- The **range** is every y that can **actually come out**. (outputs, the vertical direction)

Simple case first. For $f(x) = \sqrt{x}$, try a few inputs: $f(9) = 3$, $f(0) = 0$, but $f(-4)$ is not a real number. So the domain is $x \geq 0$. And since a square root is never negative, the outputs are $y \geq 0$. Testing a few easy numbers is always a good way to start.

1 Three ways to write a set of numbers

In words	Inequality	Interval notation	Set-builder
from -2 to 5, including -2, not 5	$-2 \leq x < 5$	$[-2, 5)$	$\{x \mid -2 \leq x < 5\}$
greater than 3	$x > 3$	$(3, \infty)$	$\{x \mid x > 3\}$
at most -1	$x \leq -1$	$(-\infty, -1]$	$\{x \mid x \leq -1\}$
all real numbers	—	$(-\infty, \infty)$	$\mathbb{R}$
every number except 2	$x \neq 2$	$(-\infty, 2) \cup (2, \infty)$	$\{x \mid x \neq 2\}$

Square bracket [ ]: the endpoint is **included** (closed dot •, $\leq$ or $\geq$).

Parenthesis ( ): the endpoint is **not included** (open dot ∘, $<$ or $>$).

∞ is not a number you can reach, so it **always** gets a parenthesis.

$\cup$ means "or" — it joins two pieces.

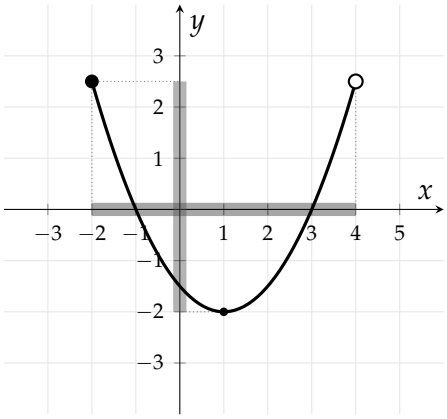
2 From points, a table, or a mapping

Example 1

$\{(-1, 4),\ (0, 2),\ (3, 4),\ (5, -1)\}$ Domain: list the x's: $\{-1, 0, 3, 5\}$. Range: list the y's: $\{-1, 2, 4\}$.

List each value **once**, from smallest to largest (4 appears twice but is written once).

3 From a graph: look at the shadows



Shine a light from above: the graph's **shadow on the x-axis** (teal) is the **domain**. Shine it from the side: the **shadow on the y-axis** (gold) is the **range**.

Read the domain left to right: from $x = -2$ (closed dot, included) to $x = 4$ (open dot, not included):

$$\text{Domain} = [-2, 4).$$

Read the range bottom to top: the lowest point is the vertex, $y = -2$. The highest y is 2.5 — the open dot does *not* reach it, but the closed dot at $(-2, 2.5)$ does:

$$\text{Range} = [-2, 2.5].$$

Watch out: an **arrow** means the graph keeps going, so that side goes to ∞ or $-\infty$. An open dot only leaves out that *one point* — check whether another part of the graph reaches the same y.

4 From an equation: what could go wrong?

Domain: start with all real numbers, then remove only what breaks

1. Division by zero	a denominator can't be 0	$\dfrac{1}{x-3}$: $x \neq 3$
2. Even roots of negatives	what is under $\sqrt{\ }$ must be ≥ 0	$\sqrt{x+5}$: $x \geq -5$
3. Real-world limits	time, length, people can't be negative	$t \geq 0$, n a whole number

Nothing else breaks: polynomials, absolute value and **odd** roots like $\sqrt[3]{x}$ have domain $\mathbb{R}$.

Range: where can the outputs go? Use the shape of the graph.

Function	Look at…	Range
$a(x-h)^2 + k$	the vertex (h, k): up if $a > 0$, down if $a < 0$	$[k, \infty)$ or $(-\infty, k]$
$a\sqrt{x-h} + k$	the starting point (h, k)	$[k, \infty)$ if $a > 0$, $(-\infty, k]$ if $a < 0$
$a\lvert x-h\rvert + k$	the vertex (h, k)	$[k, \infty)$ or $(-\infty, k]$
$\dfrac{a}{x-h} + k$	the horizontal asymptote $y = k$	every y except k
$mx + b$, $m \neq 0$	—	$\mathbb{R}$

Example 2 · a quadratic in standard form

$f(x) = -x^2 + 6x - 5$. Domain: a polynomial, so $\mathbb{R}$.
Vertex: $x = -\frac{b}{2a} = -\frac{6}{-2} = 3$, and $f(3) = -9 + 18 - 5 = 4$. It opens down ($a < 0$), so the highest output is 4: Range $= (-\infty, 4]$.

Example 3 · a square root

$g(x) = \sqrt{2x-6}+1$. Domain: $2x-6 \geq 0 \Rightarrow x \geq 3$, so $[3,\infty)$.
Range: the square root starts at 0 and grows, so g starts at $0+1=1$ and grows: $[1,\infty)$. Simple case check: $g(3)=1$, $g(5)=3$. ✓

Example 4 · a rational function — solve for x to find the range

$h(x) = \dfrac{2x+1}{x-4}$. Domain: $x \neq 4$.
Range: set $y = \dfrac{2x+1}{x-4}$ and solve for x: $yx-4y = 2x+1 \Rightarrow x(y-2) = 4y+1 \Rightarrow x = \dfrac{4y+1}{y-2}$.
This works for every y **except** $y=2$. So Range $= \{y \mid y \neq 2\}$ — the horizontal asymptote.

Common mistakes: (1) Writing the domain of $\sqrt{x-4}$ as $x > 4$ — it is $x \geq 4$, because $\sqrt{0}=0$ is fine. (2) Saying the domain of $\frac{1}{x^2+1}$ excludes something — x^2+1 is never 0, so the domain is $\mathbb{R}$. (3) Mixing up domain (x) and range (y).

5 Practice

Write every answer in interval notation (or as a list for sets of points). Test one or two easy numbers to check.

A. Sets, tables and notation EASY

1. Find the domain and range: $\{(-2,3),\ (0,5),\ (1,3),\ (4,-1)\}$

2. Find the domain and range from the table. Is it a function?

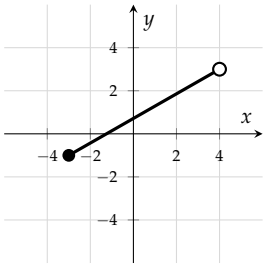
x	-3	-1	0	2	5
y	7	7	1	-2	0

3. Write in interval notation: (a) $-4 \le x < 1$ (b) $x > -2$ (c) $x \le 0$ or $x > 3$ (d) all reals except -5

B. From a graph MEDIUM

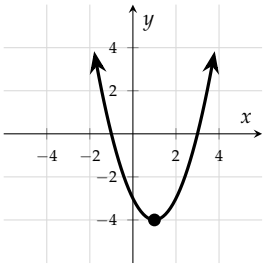
Give the domain and range of each graph. Arrows mean the graph continues.

4.



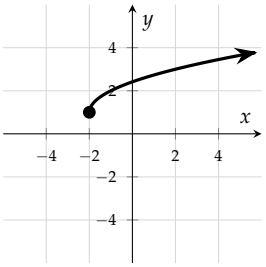
$D =$ _____ $R =$ _____

5.



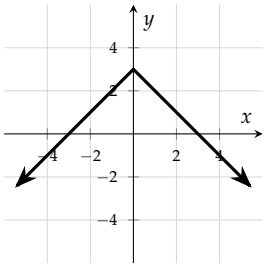
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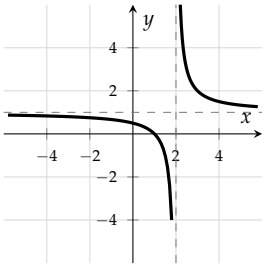
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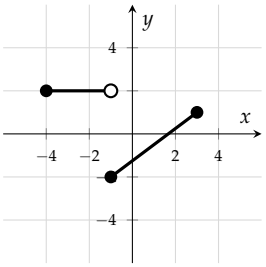
$D =$ _____ $R =$ _____

8.



$D =$ _____ $R =$ _____

9.



$D =$ _____ $R =$ _____

C. Domain from an equation MEDIUM

10. $f(x) = 3x^2 - 5x + 1$

11. $g(x) = \sqrt{x-6}$

12. $h(x) = \sqrt{10-2x}$

13. $k(x) = \dfrac{x+1}{x-7}$

14. $m(x) = \dfrac{5}{x^2-9}$

15. $p(x) = \sqrt[3]{x-4}$

16. $q(x) = \dfrac{\sqrt{x+2}}{x-3}$

17. $r(x) = \dfrac{4}{x^2+1}$

D. Domain and range from an equation MEDIUM HARD

18. MEDIUM $f(x) = (x-3)^2 + 2$

19. MEDIUM $f(x) = -2x^2 + 8x - 1$

20. MEDIUM $f(x) = \sqrt{x+4} - 3$

21. HARD $f(x) = -\sqrt{9-x} + 5$

22. MEDIUM $f(x) = |x + 1| - 4$

23. MEDIUM $f(x) = \dfrac{2}{x+3} - 1$

24. HARD $f(x) = \dfrac{3x-1}{x+2}$ (find the range by solving for x)

E. Real-world domain and range MEDIUM

25. Movie tickets cost \$12 each. A group of at most 8 friends buys n tickets, and the cost is $C(n) = 12n$. Give a reasonable domain and range.

26. A ball is thrown upward. Its height in feet after t seconds is $h(t) = -16t^2 + 64t$, until it hits the ground. Find the domain and range.

F. Challenge HARD

27. $f(x) = \sqrt{x^2 - 9}$

28. $f(x) = \dfrac{x^2 - 4}{x - 2}$

29. $f(x) = \sqrt{16 - x^2}$ (Hint: what shape is the graph?)

Before you hand it in: Did you use [ ] for included endpoints and ( ) for ∞ and open dots? Did you test one easy number in each domain?